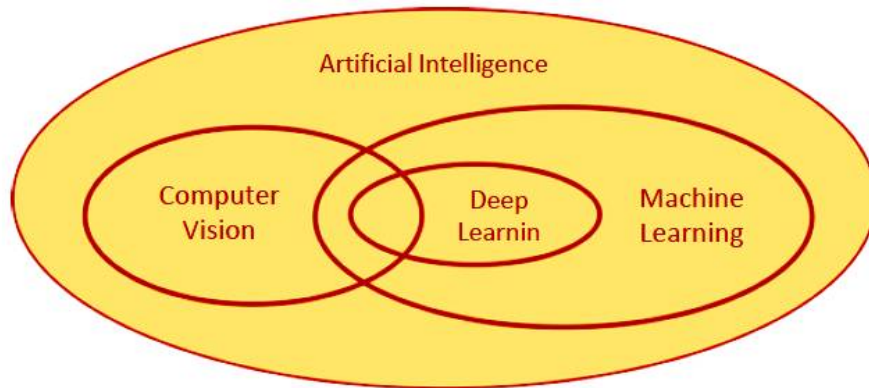


REVISION NOTES

Introduction to Computer Vision

- **Definition:** Computer Vision (CV) is a field within Artificial Intelligence (AI) that enables computers and systems to derive meaningful information from digital images, videos, and other visual inputs. It allows machines to process and analyze visual data to simulate human sight. By using algorithms and machine learning models, CV applications can detect objects, recognize patterns, and make decisions based on the visual input provided.



- **Example – Emoji Scavenger Hunt:** Imagine playing a game where a machine shows you an emoji and asks you to find a real-life object that matches it. In the “Emoji Scavenger Hunt” game, the computer uses its “vision” to detect the objects you show in front of your camera and check if they match the emoji. This simulates how CV enables machines to identify objects from real-world environments using camera input.
- **How It Works:** Computer Vision uses advanced algorithms to interpret visual data. It breaks down images into pixels, processes them using machine learning techniques, and identifies patterns, shapes, or objects by comparing them with its dataset.

Computer Vision vs. Image Processing

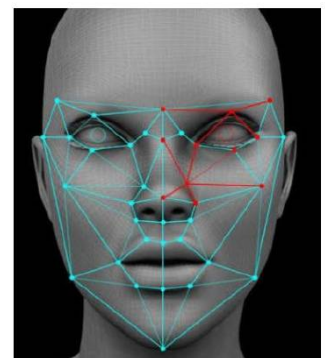
Feature	Computer Vision	Image Processing
Main Goal	To extract meaningful information from images/videos and understand them to make predictions.	To process raw input images to enhance them or prepare them for other tasks.
Scope	A superset of Image Processing.	A subset of Computer Vision.
Examples	Object detection, handwriting recognition.	Rescaling images, correcting brightness, changing tones.

Applications of Computer Vision

Over the years, CV has evolved to become a crucial part of various industries, with applications that have transformed sectors ranging from retail to healthcare. Here are some real-world applications of CV:

a. Facial Recognition

- **Definition:** Facial recognition systems identify or verify a person’s identity using their facial features.
- **Applications:**
 - **Smart Homes & Cities:** CV plays a critical role in enhancing security. In smart homes, facial recognition technology can be used to control access, allowing only

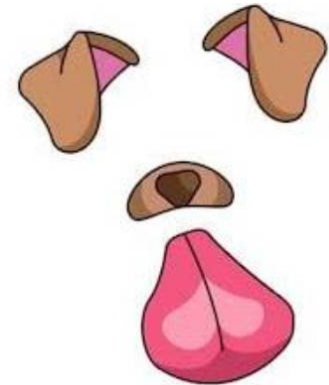


registered individuals inside. Similarly, smart city cameras can recognize and track people in public spaces for security purposes.

- **Attendance Systems:** Schools and workplaces use facial recognition for automated attendance marking.
- **Example:** Schools can track student attendance automatically by scanning students' faces upon entry.

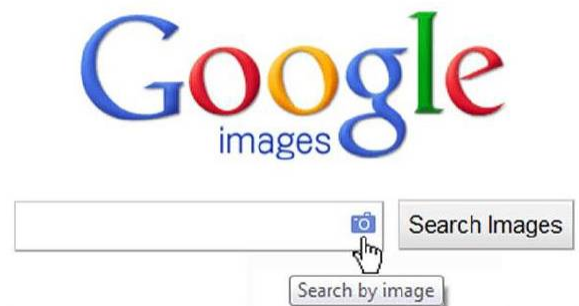
b. Face Filters in Social Media

- **Definition:** Face filters are used to apply augmented reality (AR) effects to users' faces in apps like Instagram and Snapchat.
- **How It Works:** Computer vision algorithms detect and map facial features in real-time. Using this data, the system overlays digital filters that enhance or alter the appearance of the face.
- **Example:** When you apply a dog filter on Snapchat, CV algorithms track the eyes, mouth, and nose, allowing the filter to adjust dynamically as you move.



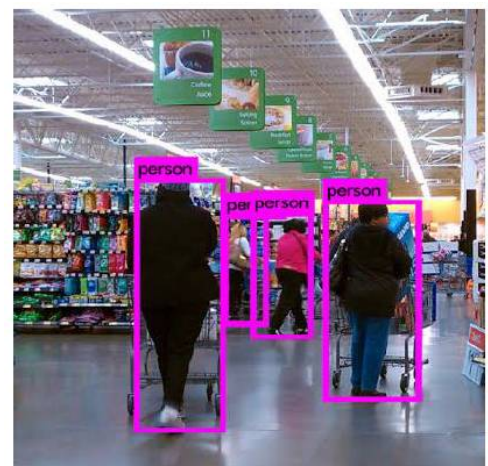
c. Google's Search by Image

- **Definition:** Google's "Search by Image" feature uses computer vision to allow users to upload an image instead of typing keywords, and Google returns relevant search results based on that image.
- **How It Works:** The CV system analyzes features like colors, shapes, and patterns of the uploaded image, compares them to images in its database, and displays matching results.
- **Example:** If you upload a picture of a landmark, Google will identify it and provide detailed information about the place, including its history and location.



d. Computer Vision in Retail

- **Customer Behavior Tracking:** Retailers use CV to track customers' movements within stores. Cameras and CV algorithms analyze how people navigate through aisles, which helps in optimizing store layouts.
- **Inventory Management:** Cameras monitor stock levels on shelves, and CV algorithms provide real-time analysis of which products need restocking.
- **Example:** Amazon Go stores use computer vision to create a cashier-less shopping experience. Shoppers can pick items off the shelf, and CV systems automatically detect what they've selected, charge their account, and let them walk out without checking out manually.



e. Self-Driving Cars

- **Definition:** Autonomous vehicles rely heavily on computer vision to interpret the surrounding environment, helping the car navigate safely without human intervention.
- **Key Tasks:** CV enables self-driving cars to detect objects like other cars, pedestrians, road signs, and obstacles. It also assists in lane detection and route navigation.
- **Example:** Tesla's autopilot uses computer vision to detect nearby vehicles and ensure lane accuracy, adjust speed, and manage traffic conditions.



f. Medical Imaging

- **Definition:** CV is revolutionizing healthcare by aiding in the analysis of medical images such as X-rays, MRIs, and CT scans.
- **How It Works:** The technology helps to identify abnormalities and diseases by converting 2D scans into detailed 3D models, offering better insights for diagnosis.
- **Example:** AI-powered systems can detect tumors or fractures from medical images faster and sometimes more accurately than human radiologists, providing early diagnosis and better treatment outcomes.



g. Google Translate App (Augmented Reality)

- **Definition:** By using CV combined with augmented reality (AR), Google Translate allows users to point their phone cameras at foreign text and receive a real-time translation overlay.
- **How It Works:** Optical character recognition (OCR) detects the foreign words, while AR translates and overlays the text in the user's preferred language.
- **Example:** If you're traveling abroad and come across a sign in a language you don't understand, pointing your camera at it will display the translated text almost instantly on your screen.

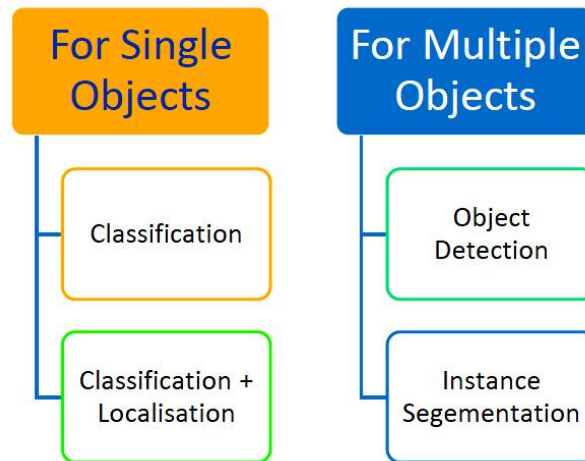


COMPUTER VISION TASKS

The tasks in computer vision are performed to extract information from an input image.

- **For Single Objects:**
 - o **Classification:** Assigning an input image a single label from a fixed set of categories.
 - o **Classification + Localization:** Identifying both what object is present and its location within the image. This is used only for single objects.
- **For Multiple Objects:**

- o **Object Detection:** Finding instances of real-world objects in images or videos, such as faces or bicycles. It involves both classification and localization for multiple objects.
- o **Instance Segmentation:** Detecting objects, giving them a category, and then assigning a label to each pixel based on that category.



BASICS OF IMAGES IN COMPUTER VISION

- **Pixels:** The smallest unit of information that makes up a digital picture. They are arranged in a 2-dimensional grid and represent the image. The more pixels an image has, the more it resembles the original.
- **Resolution:** The number of pixels in an image. It can be expressed as width by height (e.g., 1280×1024) or as a single number in megapixels (a megapixel is a million pixels).
- **Pixel Value:** A value that describes a pixel's brightness and/or color.
 - o In a byte image, the value is an 8-bit integer, giving a range of 0 to 255.
 - o 0 is typically black (no color) and 255 is white (full color).
- **Grayscale Images:** Images with shades of gray, from black (0) to white (255). Each pixel is one byte and the image has a single 2D array of pixels.
- **RGB Images:** Colored images made up of three primary colors: Red, Green, and Blue.
 - o Each pixel in an RGB image has a set of three values, one for each color channel (R, G, B), ranging from 0 to 255.
 - o These three values combine to form the complete color of the pixel.

Image Features & Convolution

- **Definition:** Features are essential visual elements in an image that help in recognizing or categorizing objects.
- **Key Features:**
 - **Edges:** Boundaries between different regions in an image.
 - **Corners:** Points where two edges meet.
 - **Blobs:** Regions that differ in properties such as color or intensity from surrounding areas.
- **Example:** In facial recognition, detecting key features like eyes, nose, and mouth edges is crucial for identification.

Convolution: A mathematical operation that is fundamental to many image processing operators. It involves the element-wise multiplication of an image array and another array called the

Kernel: A matrix that is slid across an image and multiplied with the input to enhance the output in a desirable way, such as applying filters. Convolution is used in Convolutional Neural Networks (CNNs) to extract image features.

Convolutional Neural Networks (CNN)

A **Convolutional Neural Network (CNN)** is a Deep Learning algorithm which can take in an input image, assign importance (learnable weights and biases) to various aspects/objects in the image and be able to differentiate one from the other.

CNNs are a specialized class of deep neural networks designed to process and analyze visual data. They are highly effective in tasks such as image classification, object detection, and image segmentation.

- **Structure of a CNN:**

1. **Convolution Layer:** The first layer in a CNN where filters (kernels) scan the input image to extract features like edges, colors, and textures.
 - *Example:* If you input a picture of a cat, the convolution layer extracts features like the shape of the cat's eyes, ears, and fur pattern.
2. **ReLU Layer (Rectified Linear Unit):** This activation function removes negative values from the feature maps, introducing non-linearity.
3. **Pooling Layer:** Reduces the dimensionality of the feature maps by selecting the most important information (e.g., through Max Pooling).
 - *Example:* Max Pooling selects the brightest or most prominent feature in a given region, allowing the network to focus on key details.
4. **Fully Connected Layer:** Flattens the input and uses it for classification. The flattened vector is used to assign labels to the input image.
 - *Example:* After feature extraction, the fully connected layer identifies whether the input image is a cat or a dog based on the probability distribution across labels.

- **Convolution Operation:** Convolution is the core operation in CNNs. A small matrix (kernel) is slid across the image and multiplies pixel values to detect features such as edges. The convolution output is a feature map, highlighting specific patterns in the image.
 - **Example:** Applying an edge-detection filter on an image will highlight the boundaries of objects, such as outlining a building's edges in a photograph.
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